

Tairan Song

Email: songtail@msu.edu Tel: (+1) 734-819-0577 Website: amingsong.github.io

Education

Michigan State University, East Lansing, USA Aug 2023 – Present

Ph.D. Student in Computational Mathematics, Science and Engineering

Advisor: Dr. Adam Alessio

King's College London, London, United Kingdom Sep 2021 – Sep 2022

M.Sc. in Applied Statistical Modelling & Health Informatics

Advisors: Dr. Ioannis Bakolis, Dr. Amy Ronaldson; **Graduated with Distinction**

University of Liverpool, Liverpool, United Kingdom Sep 2018 – Jun 2021

B.Sc. in Mathematics and Statistics; **Second Class Honours (Upper Division)**

Publications

Zhang, T., Jia, H., **Song, T.**, Lv, L., Gulhan, D. C., Wang, H., ... & Shen, N. (2023). De novo identification of expressed cancer somatic mutations from single-cell RNA sequencing data. *Genome Medicine*, 15(1), 115.

Conference Submissions

Song, T., Noyes, S. L., Muterspaugh, R., Munavar Ali, M. A., Zhou, S., Tobert, C., Lane, B. R., Lim, E., Alessio, A. (2025). *Predicting Kidney Cancer Grade from Highly Variable Contrast Timing CT*. Submitted to SIIM 2026, under review.

Research Experience

Spatially Aware Modeling of Renal Tumor Heterogeneity from Abdominal CT

Advisor: Dr. Adam Alessio

Michigan State University

Jan 2026 – Present

- Extending prior contrast-timing-robust radiomics frameworks toward spatially aware modeling of renal tumors, focusing on intra-tumoral and peri-tumoral heterogeneity in abdominal CT.
- Investigating sub-regional representations of renal anatomy and tumor structure to capture biologically meaningful variations beyond global tumor descriptors.
- Exploring multimodal feature fusion strategies that integrate image-derived representations with clinical variables, with an emphasis on robustness, interpretability, and cross-cohort generalization.
- Developing comparative baselines using representation learning approaches, including 3D convolutional neural networks, to benchmark against radiomics-based methods.

Kidney Cancer Grade from Highly Variable Contrast Timing CT

Advisor: Dr. Adam Alessio

Michigan State University

Jan 2025 – Dec 2025

- Developed a radiomics-based modeling framework for renal mass grading from heterogeneous abdominal CT with highly variable contrast timing, addressing protocol-induced intensity shifts beyond phase-specific imaging pipelines.

- Designed a reference-region-aware feature representation that augments tumor radiomics with quantitative descriptors from surrounding anatomical regions (normal kidney, aorta, inferior vena cava), enabling contrast-timing-robust classification without explicit phase normalization or labeling.
- Implemented and evaluated an end-to-end pipeline including automatic multi-organ segmentation (nnU-Net, TotalSegmentator), high-dimensional radiomics extraction, supervised feature reduction, and cross-validated classification, demonstrating consistent performance gains over tumor-only models for multi-class and binary grading tasks.

De Novo Identification of Expressed Cancer Somatic Mutations from scRNA Sequencing Data

Advisor: Dr. Ning Shen

Zhejiang University

Sep 2022 – Mar 2023

- Contributed to the development and evaluation of the RESA (Recurrently Expressed SNV Analysis) computational framework for high-precision de novo identification of expressed somatic mutations from scRNA-seq data, which consistently outperforms existing methods across multiple benchmark datasets.
- Implemented and validated the joint logistic regression component (RESA-jLR) within the RESA pipeline to expand the pool of detectable somatic variants and improve mutation calling performance.
- Performed extensive data preprocessing, mutation calling workflows, and empirical performance evaluation, and generated publication-ready figures supporting sensitivity and precision benchmarking against other methods reported in the study.

Teaching Experience

Teaching Assistant

Michigan State University

- CMSE 202: Computational Modeling and Data Analysis II

Jan 2025 – May 2025

Professional Experience

Research Assistant

Zhejiang University

Sep 2022 – Mar 2023

- Contributed to the RESA project for de novo identification of expressed somatic mutations from single-cell RNA sequencing data, supporting sequencing data analysis and model validation efforts.
- Provided sequencing data analysis support for collaborative mouse studies, including single-cell RNA sequencing and bulk RNA-seq experiments.

Awards

NSF NRT-IMPACTS Fellowship

Michigan State University

Awarded Sep 2023

One-semester fellowship funded by the National Science Foundation Research Traineeship Program (NSF DGE-1828149).

Skills

Programming: Python, R, SQL

ML & AI: Scikit-learn, PyTorch, TensorFlow, NLP, Transformers

MLOps & Systems: SLURM, Git, High Performance Computing, CI/CD

Research: Statistical Analysis, Translational Science, Interdisciplinary Research, Biomedical Image Analysis